

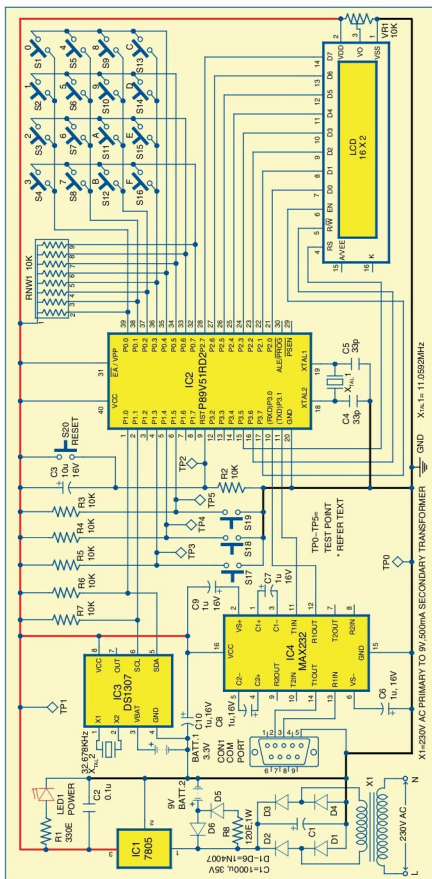


Fig. 1: Circuit of RTC-based event logger

will indicate a start or stop condition.

Start and stop. The start condition is a high-to-low transition of SDA line while SCL line is high. The stop condition is a low-to-high transition of SDA line while SCL line is high.

Acknowledge. All addresses and data words are serially transmitted to and from the RTC chip in 8-bit words. The RTC sends a zero to acknowledge that it has received each word. This happens during the ninth clock cycle.

For example, to set the time '8:32:59' into the RTC, the following interface protocol is followed:

1. Start bit is sent from microcontroller IC2 (master) to RTC IC3 (slave).

2. Then 0xD0 is sent to the slave. This is the address of the RTC (as mentioned in its datasheet). Also, the least significant bit (LSB) of 0xD0 is zero, which means it is a writing command for the slave. The slave sends an acknowledgement (ACK) after receiving this byte.

3. 0x00 is sent to the slave. It is the starting address in the RTC on which we want to write the first byte. The slave sends an ACK after receiving this address byte.

4. 0x59 (value of seconds) is sent to the slave. This is the value we want to save in the RTC on 0x00 address. The slave sends an ACK after receiving this byte.

5. 0x32 (value of minutes) is sent to the slave. This is the value we want to save in the RTC on 0x01 address. The slave sends an ACK after receiving this byte.

6. 0x08 ('8' is the value of hours, while '0' means 24-hour time format) is sent to the slave. This is the value we want to save in the RTC on 0x02

TABLE II
Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	High when S20 pressed
TP3	Low when S17 pressed
TP4	Low when S18 pressed
TP5	Low when S19 pressed